ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** making new individuals	Continuity.
2	**(d)** ovary	After fertilisation.
3	**(b)** testes	Male.
4	**(d)** binary fission	Two parts.
5	**(d)** pollination	Pollination.
6	**(a)** ovaries	Female.
7	**(d)** budding	Buds.
8	**(d)** fertilisation	Zygote.
9	**(c)** oviduct	Fallopian tube.
10	**(a)** spores	Spores.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Production of new individuals of the same kind (2).	2
12	Transfer of pollen from anther to stigma (2).	2
13	Testes (1); Ovaries (1).	1+1
14	Asexual (1); Sexual (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fission: Amoeba (1); Budding: Hydra (1); Spores: bread mould (1).	1+1+1
16	Testes (1); Vas deferens (1); Penis (1).	1+1+1
17	Small (1); Light pollen (1); Feathery stigma (1).	1+1+1
	OR: Sepals (1); Petals (1); Stamens and carpels (1).	1+1+1
18	Ovary (1); Oviduct (1); Uterus (1).	1+1+1
	OR: Place (1); Number of eggs (1); Examples (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Definition (1); Fission (1); Budding (1); Spores (1); Vegetative (1).	1+1+1+1+1
	OR: Barrier (1); Chemical (1); IUD (1); Surgical (1); Uses (1).	1+1+1+1+1
20	Sepals (1); Petals (1); Stamen (1); Carpel (1); Functions (1).	1+1+1+1+1
	OR: Cutting (1); Layering (1); Grafting (1); Tuber (1); Advantages (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that budding is the method of *Amoeba*; *Amoeba* divides by binary fission, whereas yeast and *Hydra* reproduce by budding.
- Confusing pollination (transfer of pollen) with fertilisation (fusion of male and female gametes).
- Saying that fertilisation takes place in the uterus; it takes place in the fallopian tube, and the embryo implants in the uterus.
- Writing that offspring of asexual reproduction show variation; they are identical to the parent (variation arises in sexual reproduction).
- Giving 46 as the number of chromosomes in a human sperm or egg; gametes have 23, the zygote has 46.
- Thinking the sex of the baby is decided by the mother; every egg carries X, so the type of sperm (X or Y) decides.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** nectar	Food.
2	**(c)** uterus	Womb.
3	**(a)** eyes of tubers	Vegetative.
4	**(a)** light pollen	Dry.
5	**(d)** food and oxygen	Exchange.
6	**(d)** a new plant	Vegetative.
7	**(d)** water	External fertilisation.
8	**(b)** 28 days	Monthly.
9	**(a)** stamen	Pollen.
10	**(a)** shell	Internal fertilisation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Fusion of male and female gametes (2).	2
12	Monthly discharge of blood and uterine lining (2).	2
13	Division of a cell into two equal parts (2).	2
14	Reproductive cell: sperm or ovum (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Parents (1); Gametes (1); Variation (1).	1+1+1
16	Condom (1); Pills (1); Surgery (1).	1+1+1
17	Hen (1); Frog (1); Fish (1).	1+1+1
	OR: Wind (1); Insects (1); Water (1).	1+1+1
18	Water (1); Air (1); Suitable temperature (1).	1+1+1
	OR: Human (1); Cow (1); Dog (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pollination (1); Types (1); Agents (1); Fertilisation (1); Seed and fruit (1).	1+1+1+1+1
	OR: Parents (1); Gametes (1); Variation (1); Speed (1); Examples (1).	1+1+1+1+1
20	Testes (1); Vas deferens (1); Glands (1); Urethra (1); Penis (1).	1+1+1+1+1
	OR: Water (1); Air (1); Temperature (1); Radicle (1); Plumule (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that budding is the method of *Amoeba*; *Amoeba* divides by binary fission, whereas yeast and *Hydra* reproduce by budding.
- Confusing pollination (transfer of pollen) with fertilisation (fusion of male and female gametes).
- Saying that fertilisation takes place in the uterus; it takes place in the fallopian tube, and the embryo implants in the uterus.
- Writing that offspring of asexual reproduction show variation; they are identical to the parent (variation arises in sexual reproduction).
- Giving 46 as the number of chromosomes in a human sperm or egg; gametes have 23, the zygote has 46.
- Thinking the sex of the baby is decided by the mother; every egg carries X, so the type of sperm (X or Y) decides.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 9 months	Gestation.
2	**(c)** carpel	Ovary.
3	**(d)** live young	Mammal.
4	**(b)** barrier method	Contraceptive.
5	**(a)** anther	Stamen.
6	**(b)** male gamete	Testes.
7	**(a)** 46	Pairs: 23.
8	**(d)** ovules	After fertilisation.
9	**(b)** female gamete	Ovary.
10	**(c)** 23	Half.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Condom (1); Pills (1).	1+1
12	A new individual grows from a bud on the parent (2).	2
13	Cell formed by the fusion of gametes (2).	2
14	Tiny reproductive structure in asexual reproduction (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cutting (1); Layering (1); Grafting (1).	1+1+1
16	Source of pollen (1); Variation (1); Examples (1).	1+1+1
17	Zygote (1); Embryo (1); Foetus (1).	1+1+1
	OR: Food (1); Oxygen (1); Removes waste (1).	1+1+1
18	Bright petals (1); Fragrance (1); Nectar (1).	1+1+1
	OR: Fast (1); Same as parent (1); Seedless plants (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ovary (1); Oviduct (1); Uterus (1); Vagina (1); Function (1).	1+1+1+1+1
	OR: Zygote (1); Division (1); Implantation (1); Placenta (1); Birth (1).	1+1+1+1+1
20	Ovulation (1); Lining (1); Fertilisation (1); Menstruation (1); 28 days (1).	1+1+1+1+1
	OR: Frog external (1); Hen internal (1); Eggs (1); Development (1); Care (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that budding is the method of *Amoeba*; *Amoeba* divides by binary fission, whereas yeast and *Hydra* reproduce by budding.
- Confusing pollination (transfer of pollen) with fertilisation (fusion of male and female gametes).
- Saying that fertilisation takes place in the uterus; it takes place in the fallopian tube, and the embryo implants in the uterus.
- Writing that offspring of asexual reproduction show variation; they are identical to the parent (variation arises in sexual reproduction).
- Giving 46 as the number of chromosomes in a human sperm or egg; gametes have 23, the zygote has 46.
- Thinking the sex of the baby is decided by the mother; every egg carries X, so the type of sperm (X or Y) decides.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.